

The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma

IV. Local Treatment Failure and Enucleation in the First 5 Years after Brachytherapy. COMS Report No. 19

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Objective: To describe the frequency and predictors of local treatment failure and enucleation after iodine 125 (¹²⁵) brachytherapy in patients with choroidal melanoma treated and followed up in a large randomized clinical trial.

Design: Prospective, noncomparative, interventional case series within a randomized, multicenter clinical trial.

Participants: Patients enrolled in the Collaborative Ocular Melanoma Study (COMS) trial of enucleation versus brachytherapy between February 1987 and July 1998; tumors measured 2.5 to 10.0 mm in apical height and no more than 16.0 mm in longest basal dimension.

Methods: ¹²⁵ brachytherapy was administered via episcleral plaque according to a standard protocol. Follow-up ophthalmic evaluations, including ophthalmic ultrasound and fundus photography, were performed according to a standard protocol at baseline, every 6 months thereafter for 5 years, and subsequently at annual intervals. Survival analysis methods were used to estimate the cumulative risk of postirradiation treatment failure and enucleation. Factors associated with treatment failure and enucleation of plaqued eyes were evaluated using Cox proportional hazards analysis.

Main Outcome Measures: Reports of enucleation and of local treatment failure, defined as tumor growth, recurrence, or extrascleral extension, derived from clinical reports based on echographic and photographic documentation.

Results: As of September 30, 2000, 638 of the 650 patients randomized to brachytherapy and so treated had been followed up for 1 year or longer, and 411 had been followed up for at least 5 years. Sixty-nine eyes were enucleated during the first 5 years after brachytherapy, and treatment failure was reported for 57 eyes. The Kaplan-Meier estimate of proportion of patients undergoing enucleation by 5 years was 12.5% (95% confidence interval [CI], 10.0%–15.6%); the risk of treatment failure was 10.3% (95% CI, 8.0%–13.2%). Treatment failure was the most common reason for enucleation within 3 years of treatment; beyond 3 years, ocular pain was most common. Risk factors for enucleation were greater tumor thickness, closer proximity of the posterior tumor border to the foveal avascular zone, and poorer baseline visual acuity in the affected eye. Risk factors for treatment failure were older age, greater tumor thickness, and proximity of the tumor to the foveal avascular zone. Local treatment failure was associated weakly with reduced survival after controlling for baseline tumor and personal characteristics (adjusted risk ratio, 1.5; *P* = 0.08).

Conclusions: Local treatment failure and enucleation were relatively infrequent events after ¹²⁵ brachytherapy within the COMS. Treatment failure typically occurred early and was associated weakly with poorer survival. The COMS randomized trial documented the absence of a clinically or statistically significant difference in survival for patients randomly assigned to enucleation versus brachytherapy. This analysis documents the efficacy of brachytherapy to achieve sustained local tumor control and to conserve the globe. *Ophthalmology* 2002;109:2197–2206 © 2002 by the American Academy of Ophthalmology.

The goal of treatment for choroidal melanoma is to preserve the patient's life through removal or destruction of the tumor, traditionally achieved by enucleation of the involved

eye or by radiation therapy.^{1,2} The Collaborative Ocular Melanoma Study (COMS) has conducted a randomized trial to evaluate the effectiveness of globe-conserving radiother-

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